

Harshit Mani Tripathi

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Summary

DevOps / Platform Engineer with production experience at Unstop, launched a user-facing cloud-native coding workspace platform. Recommended by Sr. Director of Tech and senior engineers at Unstop. Building homelab infrastructure since 2021.

Experience

DevOps Intern | Unstop, Delhi

March 2026 - June 2026

- Engineered a production-grade, user-facing EKS-based coding workspace platform with zero-downtime rollouts via Kubernetes rolling updates, serving public users at scale.
- Implemented a FastAPI microservice enabling isolated, persistent coding workspaces in a multi-tenant architecture across 3 access tiers (organizer/student/evaluator), eliminating per-tenant EC2 provisioning; designed and iterated API contracts in collaboration with the backend team.
- Reduced coding workspace EBS costs by ~80% via snapshot-based images; identified and proposed ~30-40% additional savings across AWS environment through compute right-sizing and resource cleanup.
- Implemented Terraform configurations provisioning production-grade EKS clusters with Karpenter for dynamic node autoscaling, securing public-facing infrastructure with Traefik ingress and TLS termination.
- Built custom SigNoz log preprocessors and developer-facing telemetry dashboards; authored VPA resource-tuning scripts to optimize cluster costs across staging environment.

Infrastructure Projects

Hybrid Infrastructure Lab

github.com/hudater/homelab/

- Designs and operates a cost-optimized, multi-cloud hybrid infrastructure spanning on-premise Kubernetes and OCI cloud edge workloads with cross-region high availability.
- Enforces zero-trust network isolation via OPNsense, NetBird, and VLAN segmentation across all workloads.
- Manages full lifecycle deployment via OpenTofu across OCI, k3s, and Cloudflare; infrastructure and services as code.

Recursive DNS & Observability Stack

blog.hudater.dev/posts/dns-infra/

- Operates core recursive DNS routing via Technitium and Unbound, handling ~500k daily queries with 75% cache hit rate.
- Runs Prometheus, Grafana, Loki, and Tempo for unified metrics, logs, and distributed tracing across hybrid workloads.

Open Source, Community & Publications

- Linkme Generator** | github.com/ironicbadger/linkme/pulls?q=author:hudater
Contributed 5+ features in Go and automated multi-stage CI/CD deployments via GitHub Actions.
- CNCF SRE Games** | showcase.hudater.dev/sre_games.pdf
Investigated and resolved simulated Kubernetes microservice outages under time constraints using distributed tracing.
- Technical Writing** | blog.hudater.dev
Authored in-depth deep-dives on DNS internals, core networking protocols, and Git object model.

Skills

- Cloud & Infrastructure:** AWS, Oracle Cloud Infrastructure (OCI), Proxmox, Linux, High Availability, Cloud-Native Infrastructure
- Containers & Orchestration:** Kubernetes, Docker, Helm, Karpenter
- IaC, CI/CD & Version Control:** Terraform (OpenTofu), Ansible, Git, GitHub Actions
- Languages:** Go, Python, FastAPI, Bash
- Networking & Security:** OPNsense, NetBird, WireGuard, VPC, VLAN, Traefik, Keycloak, Authentik
- Observability:** SigNoz, Prometheus, Grafana, Loki, Tempo, Goldilocks (VPA)

Education

B. Tech CSE (Data Science) | Noida International University
8/10 GPA

Aug 2024 - Aug 2028